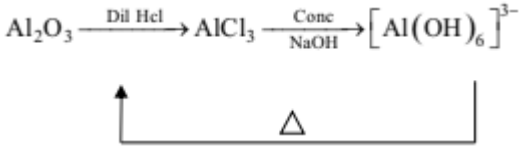
Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

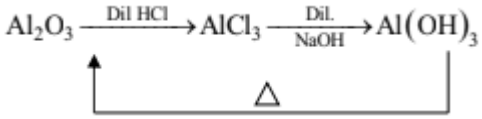
Q.1

Which of the following is correct related to change in aluminium oxide?

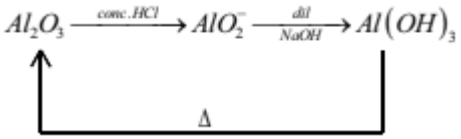
(A)



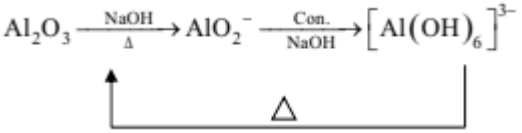
(B)



(C)



(D)



Q.2

In which of the following the central atom is not in the sp^3 hybridised state?

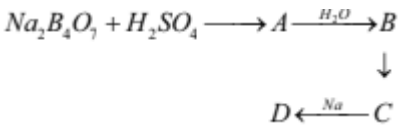
(A) Boron in solid layered boric acid

(B) Boron in diboare

(C) Boron in the product formed by the reaction of sodium hydride with diborane

(D) Boron in the product formed by the addition of $(CH_3)_3\ N$ to diborane.

Q.3



The incorrect statement regarding substance ' D ' is

(A) 'D' will not react with HCl

(B) 'D' will not react with dil H_2SO_4

@IITJEE_Advanced

(C) 'D' will react with conc. H_2SO_4 to from substance 'B'

(D) 'D' will react with conc. H_2SO_4 to from substance 'A'.

Q.4

Which of the following is correct regarding the hydrolysis of BX_3 (X-halogen)?

(A) All BX_3 undergo hydrolysis to produce $B(OH)_3(aq)$ and $HX(aq)$

(B) BF_3 does not undergo complete hydrolysis due to formation of HBF_4

(C) BBr_3 does not undergo hydrolysis at all because it cannot form hydrogen bonds with water

(D) All the above are correct

Q.5

The silicate anion in the mineral kaionite is a chain of three SiO_4 Tetrahedrons that share corners with adjacent tetrahedra. The mineral also contains Ca^{2+} ions, Cu^{2+} ions, and water molecules in a 1: $1:1$ ratio. Mineral is represented as:

(A) $CaCuSi_3O_{10} \cdot H_2O$

(B) $CaCuSi_3O_{10} \cdot 2H_2O$

(C) $Ca_2Cu_2Si_3O_{10} \cdot 2H_2O$

(D) none of these

Q.6

In the following sub questions, choose the correct answer from among the following possibilities and select correct code of your answer (Answer of 1, 2, 3 and 4 respectively)
1) The most stable low valent halide (1) $GeCl_2$ (2) $SnCl_2$ (3) $PbCl_2$.
2) A non existing halide (1) $SnCl_4$ (2) $PbCl_4$ (3) PbI_4.
3) A purely acidic oxide (1) PbO_2 (2) SnO_2 (3) SiO_2
4) Thermally most stable hydride (1) NH_3 (2) PH_3 (3) AsH_3.

(A) $3, 2, 1, 3$

(B) $1, 3, 3, 1$

(C) $3, 3, 3, 1$

(D) $1, 1, 1, 3$

Q.7

A scarlet red compound (X) on treatment with conc. HNO_3 gives compounds($\mathbf{Y}$) and (Z). (Z) with HCL produces a chloride compond (A) which can also be produced by treating (X) with conc HCL. compounds (X), (Z) and (A) will be

(A) $Mn_3O_4, MnO_2, MnCl_2$

(B) $Pb_3O_4, PbO_2, PbCl_2$

@IITJEE_Advanced

(C) $Fe_3O_4, Fe_2O_3, FeCl_2$

(D) $Fe_3O_4, Fe_2O_3, FeCl_3$

Q.8

An inorganic compound (A) made by two most occuring elements in the earth crust having polymeric tetrahedral network structure.With carbon, compound (A) produces a poisonous gas (B) which is the most stable diotomic molecule.Compound (A) and (B) will be

(A) SiO_2, CO_2

(B) SiO_2, CO

(C) SiC, CO

(D) $SiO_2,\ N_2$

Q.9

The b.p of CF_4 is almost half that of SiF_4 while the difference in b.p of CCl_4 and $SiCl_4$ is little. This is because

(A) the negative charge on fluorine causes greater repulsion between neighbouring molecules in CF_4 where as it is less in SiF_4 due to back bonding form $F \rightarrow Si$

(B) in $SiCl_4$ inter-molecular repulsions are more due to the negative charge chlorine as π back bonding formed $Cl \rightarrow Si$ is

(C) C - Cl bonds in CCl_4 are less polar then $Si - Cl$ bonds in $SiCl_4$

(D) All are correct

Q.10

Which of the following statements is/are correct with respect to the property of elements in the carbon family with an increase in atomic number, their

(A) Atomic size decreases

(B) Ionization energy increases

(C) Metallic character decreases

(D) Stability of $+2$ oxidation state increases

Q.11

Which of the following is/are incorrect statements -

(A) The hydroxide of aluminium is more acidic than that of boron

(B) The hydroxide of boron is basic, while that of aluminium is amphoteric

(C) The hydroxide of boron is acidic, while that of aluminium is amphoteric

(D) The hydroxide of boron and aluminium are amphoteric

@IITJEE_Advanced

Q.12

Passage:
Boron is not affected by non oxidizing acids where as oxidizing acids react with boron to give boric acid. Boron also dissolves in excess alkali. Eventhough aqueous solution of boric acid acts as a weak acid, it is neutral to bromocresol green indicator (pH range: $3.8 - 5.4$). But aqueous solution of potassium hydrogen fluoride is acidic to this indicator. Boron forms trihalides, which are covalent and act as Lewis acids. BF_3 forms addition product, $H_3O^+ [BF_3OH]^-$ with water.
Question: Aqueous solution containing excess boric acid in potassium hydrogen fluoride is alkaline to bromocresol green indicator. This is because of the reaction -

(A) $B(OH)_3 + KHF_2 \longrightarrow K[B(OH)_3HF_2] + KOH + H_2O$

(B) $B(OH)_3 + KHF_2 \longrightarrow K_3BO_3 + KF + KOH + H_2O$

(C) $B(OH)_3 + KHF_2 \longrightarrow KBF_4 + KOH + H_2O$

(D) $B(OH)_3 + KHF_2 \longrightarrow K[B(OH)F] + KOH + H_2O$

Q.13

Passage:
Boron is not affected by non oxidizing acids where as oxidizing acids react with boron to give boric acid. Boron also dissolves in excess alkali. Eventhough aqueous solution of boric acid acts as a weak acid, it is neutral to bromocresol green indicator (pH range: $3.8 - 5.4$). But aqueous solution of potassium hydrogen fluoride is acidic to this indicator. Boron forms trihalides, which are covalent and act as Lewis acids. BF_3 forms addition product, $H_3O^+ [BF_3OH]^-$ with water.
Question: On mixing BBr_3 with excess $(C_2H_5)_2NH$ in benzene solvent gives -

(A) $B[N(C_2H_5)_2]_3$

(B) BN

(C) $B(C_2H_5)_3$

(D) $B[N(C_2H_5)_2]Br_2$

Q.14

Passage:
Boron is not affected by non oxidizing acids where as oxidizing acids react with boron to give boric acid. Boron also dissolves in excess alkali. Eventhough aqueous solution of boric acid acts as a weak acid, it is neutral to bromocresol green indicator (pH range: $3.8 - 5.4$). But aqueous solution of potassium hydrogen fluoride is acidic to this indicator. Boron forms trihalides, which are covalent and act as Lewis acids. BF_3 forms addition product, $H_3O^+ [BF_3OH]^-$ with water.
Question: On mixing BCl_3 with excess aqueous $NaCl$ in acidic medium gives -

(A) Na_3BO_3 (aq)

(B) $NaBCl_3(OH)$ (aq)

(C) $NaBCl_4$ (aq)

(D) $B(OH)_3$ (aq)

@IITJEE_Advanced

Q.15

Passage:
Compound (A) on reduction with $LiAlH_4$ gives a hydride (P) containing 21.72% hydrogen along with other products. The one mole of hydride (P) and 2 mole of ammonia at higher temperature gives a compound (Q) which is known as inorganic benzene. (A) hydrolyses incompletely and forms a compound (R) and H_3BO_3.
Question: In hydride (P) (Select correct statement) -

(A) the central atom has trigonal planar geometry

(B) all H-atom lie in the same plane

(C) all four terminal $B-H$ bond lengths are equivalent but that of four bridging $B-H$ bond lengths are not equivalent

(D) a three-centre two-electron bond $(3c-2e)$ is formed by overlap of an sp^3 hybrid orbital from each boron atom with the 1 s orbital of hydrogen atom.

Q.16

Passage:
Compound (A) on reduction with $LiAlH_4$ gives a hydride (P) containing 21.72% hydrogen along with other products. The one mole of hydride (P) and 2 mole of ammonia at higher temperature gives a compound (Q) which is known as inorganic benzene. (A) hydrolyses incompletely and forms a compound (R) and H_3BO_3.
Question: The hybridisation of central atom of compound (R) is $-$

(A) sp^2

(B) sp^3

(C) sp

(D) sp^3 d

Q.17

Passage:
Compound (A) on reduction with $LiAlH_4$ gives a hydride (P) containing 21.72% hydrogen along with other products. The one mole of hydride (P) and 2 mole of ammonia at higher temperature gives a compound (Q) which is known as inorganic benzene. (A) hydrolyses incompletely and forms a compound (R) and H_3BO_3.
Question: Which of the following statement is incorrect for the compound (A) -

(A) It has trigonal planar geometry

(B) The bond length between the central atom and the substituent atom is shorter than the sum of the covalent radii.

(C) The coordination geometry around central atom of compound (A) and N atom in $1:1$ complex of (A) and NH_3 is same.

(D) In compound (A), there is $p\pi-d\pi$ bonding.

Q.18

@IITJEE_Advanced

Match Column I with Column II.

Column I	Column II
(A) Peroxide	(p) C_3O_2
(B) Superoxide	(q) PbO_2
(C) Dioxide	(r) KO_2
(D) Suboxide	(s) H_2O_2

(A) A-(r) B-(s) C-(p) D-(q)

(B) A-(s) B-(r) C-(p) D-(q)

(C) A-(s) B-(r) C-(q) D-(p)

(D) A-(r) B-(s) C(q) D-(p)

Q.19

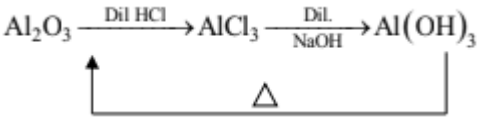
The coordination number of Al in the crystalline state of $\mathrm{AlCl_3}$ is

Q.20

The number of $\mathrm{R_2Si(OH)_2}$ units required to prepare a silicone polymer containing $10\mathrm{Si-O-Si}$ linkages is

Answers & Solutions

Q.1 Answer:



Solution:

A) $\mathrm{Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O}$

B) $\mathrm{AlCl_3 + 6NaOH \rightarrow Na_3\left[Al(OH)_3\right] \xrightarrow{\Delta} Na_3AlO_3 + 3H_2O}$

Q.2 Answer:

Boron in solid layered boric acid

Solution:

Boron in boric acid: $\mathrm{sp^2}$

Boron in diborane: $\mathrm{sp^3}$

$\mathrm{2NaH + B_2H_6 \rightarrow 2Na\left[BH_4\right]}$

Boron in $\mathrm{\left[BH_4\right]^-}$: $\mathrm{sp^3}$

$\mathrm{B_2H_6 + 2\,N\left(CH_3\right)_3 \rightarrow 2\left[H_3\,B \leftarrow N\left(CH_3\right)_3\right]}$

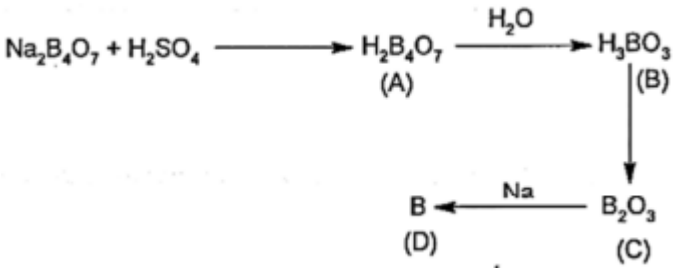
Boron in $\mathrm{\left[H_3\,B \leftarrow N\left(CH_3\right)_3\right]}$: $\mathrm{sp^3}$

@IITJEE_Advanced

Q.3 Answer:

'D' will react with conc. H_2SO_4 to from substance 'B'

Solution:



$\therefore$ Boron does not react with HCl, dil H_2SO_4 but react with conc. H_2SO_4 to form H_3BO_3

$$2\ B + \underset{\text{(cone.)}}{3H_2SO_4} \longrightarrow 2H_3BO_3 + 3SO_2$$

Q.4 Answer:

BF_3 does not undergo complete hydrolysis due to formation of HBF_4

Solution:

BF_3 hydrolyses incompletely and form fluioroborates. This is because the HF first formed reacts with the H_3BO_3.

$$4BF_3 + 12H_2O \rightarrow 4\ B(OH)_3 + 12HF$$

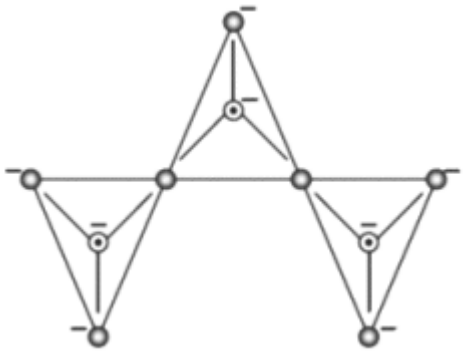
$$12HF + 3\ B(OH)_3 \rightarrow 3H\left[BF_4\right] + 9H_2O$$

$$4BF_3 + 3H_2O \rightarrow B(OH)_3 + 3H\left[BF_4\right]$$

Q.5 Answer:

$Ca_2Cu_2Si_3O_{10} \cdot 2H_2O$

Solution:



when three SiO_4 tetrahedrons are joined the silicate ion is $Si_3O_{10}^{8-}$. As the 8 negative charges are neutralized by cations Ca^{2+} and Cu^{2+} in 1:1 ratio. The formula of this silicate is $Ca_2Cu_2Si_3O_{10} \cdot 2H_2O$

Q.6 Answer:

$3, 3, 3, 1$

Solution:

@IITJEE_Advanced

Stability $Pb^{+2} > Sn^{+2} > Ge^{+2}$

in PbI_4, Pb^{+4} **is less stable**

Acidic nature :

$CO_2 > SiO_2 > GeO_2 > SnO_2 > PbO_2$

$GeO_2SnO_2PbO_2 =$ **amphoteric**

B.E. $= N - H > P - H > As - H$

Q.7 Answer:

$Pb_3O_4, PbO_2, PbCl_2$

Solution:

X is

Pb_3O_4

$Pb_3O_4 + 4HNO_3 \rightarrow Pb(NO_3)_2 + PbO_2 + 2H_2O$

$Pb(NO_3)_2 + 2HCl \rightarrow PbCl_2 + 2HNO_3$

$Pb_3O_4 + 8HCl \rightarrow 3PbCL_2 + 4H_2O + Cl_2$

Q.8 Answer:

SiO_2, CO

Solution:

The reaction of is,

$SiO_2 + 2C \rightarrow Si + 2CO$

Compound A **is silica** (SiO_2)**, which has a polymeric tetraheadral network structure and is formed of the two elements that are most frequently found in the earth's crust: silicon and oxygen.**

The most stable diatomic molecule is CO**, a toxic gas that makes up compound** B**.**

Q.9 Answer:

All are correct

Solution:

since fluorine is more electronegative their develops a negative charge on fluorine atoms in $C - F$ **bonds. But in Si-F bond the negative charge developed on** F **atoms is less due to back bonding. Due to more negative charge** F **atoms in** CF_4 **the inter-molecular repulsions are more thus have low b.p.**

Q.10 Answer:

Stability of $+2$ oxidation state increases

Solution:

In carbon family stability $+2$ **oxidation state increases on moving down the group in the periodic table with an increase in atomic number due to screening effect.**

Q.11 Answer:

@IITJEE_Advanced

The hydroxide of aluminium is more acidic than that of boron

Solution:

$B(OH)_3 \Rightarrow H_3BO_3$ **Boric acid**

$Al(OH)_3 \Rightarrow$ **Amphoteric**

Q.12 Answer:

$B(OH)_3 + KHF_2 \longrightarrow KBF_4 + KOH + H_2O$

Solution:

(i) $B(OH)_3 + KHF_2 \longrightarrow KBF_4 + KOH + H_2O$

(ii) $BBr_3 + 3\left(C_2H_5\right)_2 NH \longrightarrow B\left[N\left(C_2H_5\right)_2\right]_3 + 3HBr$

HBr **protonates excess diethyl amine**

(iii) Except BF_3 **all boron trihalides are hydrolysed in aqueous medium.**

$BCl_3(g) + 3H_2O \longrightarrow B(OH)_3(aq) + 3HCl(aq)$

Q.13 Answer:

$B\left[N\left(C_2H_5\right)_2\right]_3$

Solution:

(i) $B(OH)_3 + KHF_2 \longrightarrow KBF_4 + KOH + H_2O$

(ii) $BBr_3 + 3\left(C_2H_5\right)_2 NH \longrightarrow B\left[N\left(C_2H_5\right)_2\right]_3 + 3HBr$

HBr **protonates excess diethyl amine**

(iii) Except BF_3 **all boron trihalides are hydrolysed in aqueous medium.**

$BCl_3(g) + 3H_2O \longrightarrow B(OH)_3(aq) + 3HCl(aq)$

Q.14 Answer:

$B(OH)_3$ (aq)

Solution:

(i) $B(OH)_3 + KHF_2 \longrightarrow KBF_4 + KOH + H_2O$

(ii) $BBr_3 + 3\left(C_2H_5\right)_2 NH \longrightarrow B\left[N\left(C_2H_5\right)_2\right]_3 + 3HBr$

HBr **protonates excess diethyl amine**

(iii) Except BF_3 **all boron trihalides are hydrolysed in aqueous medium.**

$BCl_3(g) + 3H_2O \longrightarrow B(OH)_3(aq) + 3HCl(aq)$

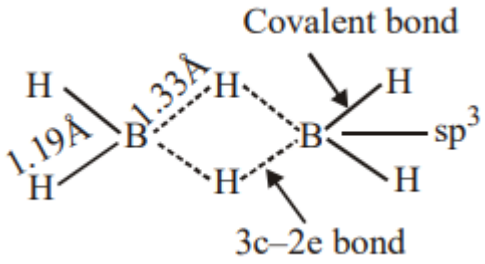
Q.15 Answer:

a three-centre two-electron bond $(3c-2e)$ is formed by overlap of an sp^3 hybrid orbital from each boron atom with the 1 s orbital of hydrogen atom.

Solution:

@IITJEE_Advanced

(i)

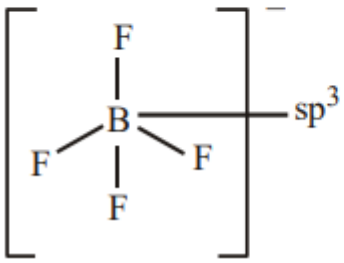


(ii)

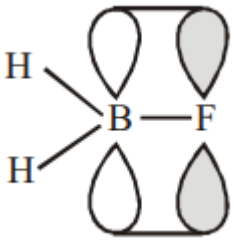
$$4BF_3 + 12H_2O \rightarrow 4H_3BO_3 + 12HF$$

$$12HF + 3H_3BO_3 \rightarrow 3H^+ + 3[BF_4]^- + 9H_2O$$

$$4BF_3 + 3H_2O \rightarrow H_3BO_3 + 3H^+ + 3[BF_4]^-$$



(iii)



There is $2p\pi - 2p\pi$ **bonding**

$4BF_3(A) + 3LiAlH_4 \xrightarrow{\text{Ether}} 2\,B_2H_6(P) + 3LiAlF_4\; B_2H_6 + 3O_2 \longrightarrow B_2O_3 + 3H_2O +$ **Heat**

Molecular weight of compound $(P) = 21.76 + 6 = 27.76$

$\%$ **of H in compound (P) i.e.,** $B_2H_6 = \frac{6}{27.76} \times 100 = 21.72$

$B_2H_6 + NH_3 \xrightarrow{\Delta} B_3\,N_3H_6$ **(inorganic benzene)** $+H_2$

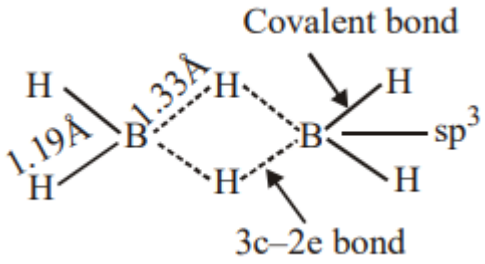
Q.16 Answer:

sp^3

Solution:

@IITJEE_Advanced

(i)

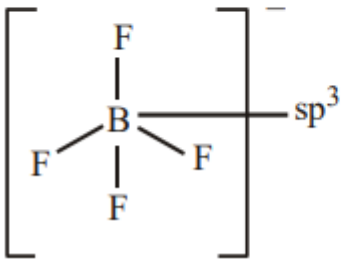


(ii)

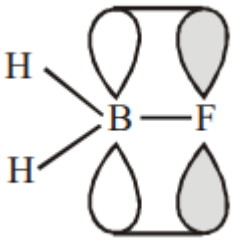
$$4BF_3 + 12H_2O \rightarrow 4H_3BO_3 + 12HF$$

$$12HF + 3H_3BO_3 \rightarrow 3H^+ + 3[BF_4]^- + 9H_2O$$

$$4BF_3 + 3H_2O \rightarrow H_3BO_3 + 3H^+ + 3[BF_4]^-$$



(iii)



There is $2p\pi - 2p\pi$ **bonding**

$4BF_3(A) + 3LiAlH_4 \xrightarrow{\text{Ether}} 2\,B_2H_6(P) + 3LiAlF_4\; B_2H_6 + 3O_2 \longrightarrow B_2O_3 + 3H_2O +$ **Heat**

Molecular weight of compound $(P) = 21.76 + 6 = 27.76$

$\%$ **of H in compound (P) i.e.,** $B_2H_6 = \frac{6}{27.76} \times 100 = 21.72$

$B_2H_6 + NH_3 \xrightarrow{\Delta} B_3\,N_3H_6$ **(inorganic benzene)** $+H_2$

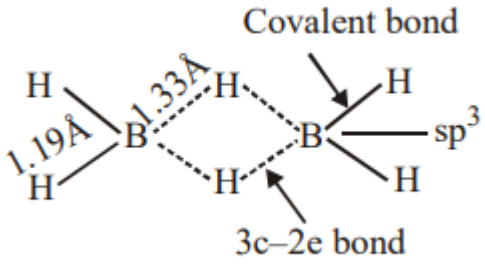
Q.17 Answer:

In compound (A), there is $p\pi - d\pi$ bonding.

Solution:

@IITJEE_Advanced

(i)

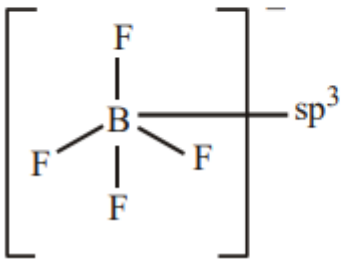


(ii)

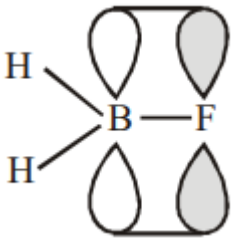
$$4BF_3 + 12H_2O \rightarrow 4H_3BO_3 + 12HF$$

$$12HF + 3H_3BO_3 \rightarrow 3H^+ + 3[BF_4]^- + 9H_2O$$

$$4BF_3 + 3H_2O \rightarrow H_3BO_3 + 3H^+ + 3[BF_4]^-$$



(iii)



There is $2p\pi - 2p\pi$ bonding

$4BF_3(A) + 3LiAlH_4 \xrightarrow{\text{Ether}} 2\,B_2H_6(P) + 3LiAlF_4\; B_2H_6 + 3O_2 \longrightarrow B_2O_3 + 3H_2O +$ **Heat**

Molecular weight of compound $(P) = 21.76 + 6 = 27.76$

% of H in compound (P) i.e., $B_2H_6 = \frac{6}{27.76} \times 100 = 21.72$

$B_2H_6 + NH_3 \xrightarrow{\Delta} B_3\,N_3H_6$ **(inorganic benzene)** $+H_2$

Q.18 Answer:

A-(s) B-(r) C-(q) D-(p)

Solution:

$(A) - \mathbf{s}, (B) - \mathbf{r}, (C) - \mathbf{q}, (D - \mathbf{p}.$

(B) Super oxide is KO_2 **(r),**

(A) Peroxide is H_2O_2 **(s),**

@IITJEE_Advanced

(D) Suboxide is C_3O_2 (p)

(C) Dioxide is PbO_2 (q);

Q.19 Answer:

—

Solution:

In the crystalline state of $AlCl_3$, the Cl^- ions form space lattice with Al^{3+} occupying octahedral voids. The coordination number of Al in the crystalline state of $AlCl_3$ is 6 .

Q.20 Answer:

—

Solution:

To prepare a silicone containing $nSi-O-Si$ linkages, the total number of units required are $(n+1)$, of which 2 are chain blocking units, R_3SiOH and $(n-1)$ are chain building units, $R_2Si(OH)_2$. Thus, the number of $R_2Si(OH)_2$ units needed are 9 .

@IITJEE_Advanced